

Arterial, Venous, and Lymphatic Disease

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Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1. Establish a general approach to the vascular patient.
2. Recognize the clinical presentation of arterial, venous, and lymphatic disorders.
3. Understand the complications of arterial, venous, and lymphatic disorders.
4. Synthesize the proper workup for patients presenting with arterial, venous, and lymphatic disease.
5. Understand and appreciate therapeutic options and algorithms in the treatment of arterial, venous, and lymphatic disorders.

Approach to the Vascular Patient – History Taking

Clinical History (*timing* is very important)

- Pain, weakness in the extremities
- Neurosensory complaints (warmth, numbness, hypersensitivity)
- Discoloration, swelling, tissue loss/ulceration, varicosities

Medical History

- Hypertension
- Atherosclerotic disease (CAD, PAD, CVA, carotid stenosis)
- Hyperlipidemia
- Diabetes mellitus

Surgical History

- Everything

Social History

- Smoking

Common or Concerning Symptoms

- Abdominal, flank, or back pain
- Pain or weakness in the arms or legs
- Intermittent claudication
- Cold, numbness, pallor in the legs; hair loss
- Swelling in calves, legs, or feet
- Color change in fingertips or toes in cold weather
- Swelling with redness or tenderness

Approach to the Vascular Patient – Physical Exam

Inspection

- Meticulous inspection and documentation of skin changes
 - Atrophy, cyanosis, ulceration, hair loss, abnormal nail growth
 - Pictures are often taken to monitor wound care

Palpation of pulses

- Unilateral: Abdominal aorta
- Bilateral: carotid, brachial, radial, ulnar, femoral, popliteal, posterior tibial, & dorsalis pedis arteries

Auscultation & Ultrasound

- Auscultate for carotid & renal bruits (indicative of turbulent flow)
- Screening ultrasound for abdominal aortic aneurysm (AAA)
- Follow-up ultrasound for:
 - Patients with known or suspected vascular disease
 - Patients following a vascular intervention

Approach to the Vascular Patient – Physical Exam

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Approach to the Vascular Patient – Pulse Exam

Femoral → Midway between ASIS and pubic tubercle

Popliteal → Popliteal fossa, knee at 45° angle, with two hands

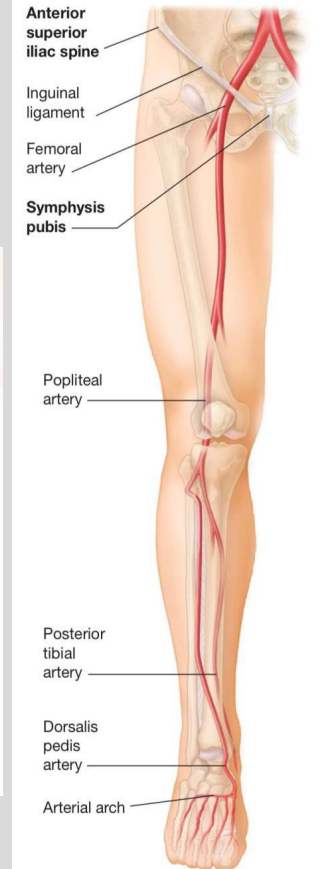
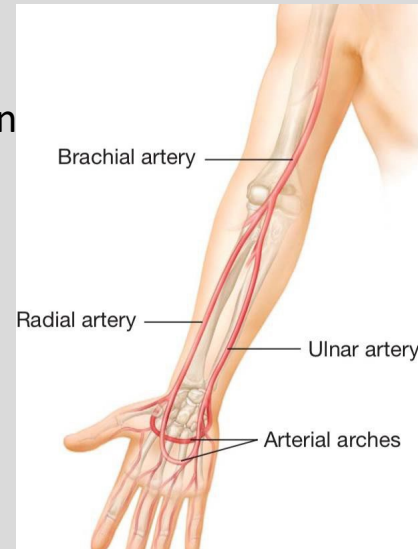
Posterior tibial → 2 cm posterior to medial malleolus

Dorsalis pedis → 1 cm lateral to extensor hallucis longus tendon

Always compare sides

Recommended Grading of Pulses

3+	Bounding
2+	Brisk, expected (normal)
1+	Diminished, weaker than expected
0	Absent, unable to palpate



Arterial Disease

Aorta (covered in another lecture)

- Aneurysm: dilation, can lead to rupture
- Dissection: intimal tear → creation of false lumen

Peripheral arteries (typically limbs)

- Peripheral artery disease (PAD)
- Aneurysm (MC = popliteal, 2nd = femoral)

Extra-cranial cerebrovascular arteries

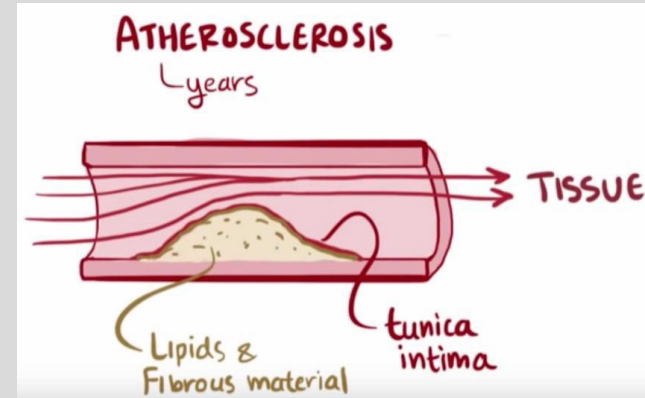
- Carotid stenosis
- Vertebral artery compression

Visceral arteries (renal, splanchnic, & mesenterics)

- Stenosis, occlusion, aneurysm, vasculitides

Peripheral Arterial Disease (PAD)

- Obstruction or deterioration of arteries *other than those supplying the brain or the heart*. Commonly affects the lower extremity.
 - Major pathophysiologic process is **atherosclerosis**
- Patients typically present with **claudication** (pain with activity, that is relieved by rest, and reproducible with **the same level of activity**) +/- a non-healing wound
- Affects 20% of Americans ≥65 years old
- Increased risk with obesity, non-white ethnicities, smoking, diabetes, HTN and HLD (same risk factors as atherosclerosis)



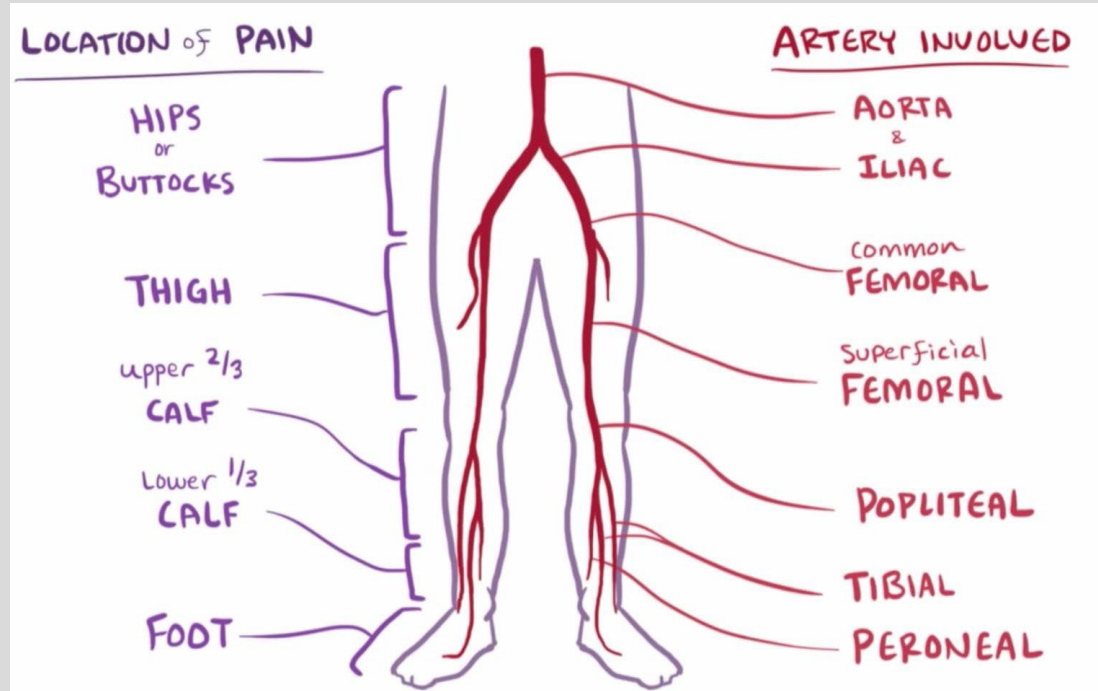


Pseudoclaudication

- Claudication vs pseudoclaudication
- **Pseudoclaudication:** Leg pain caused by nerve impingement (spinal stenosis) rather than arterial obstruction. Aka neurogenic claudication.
 - Variable walking distances
 - Positional relief (leaning forward)

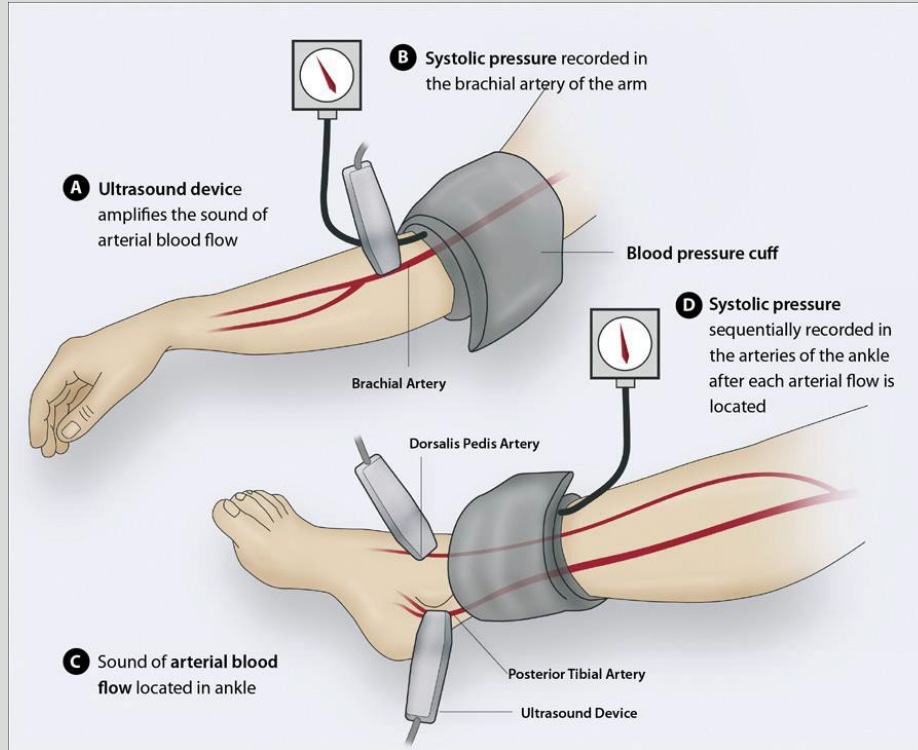


PAD – Clinical Presentation



Patients typically present with **claudication**: pain with activity, that is relieved by rest, and **reproducible** with **the same level of activity**.

Diagnosis of PAD – Ankle-Brachial Index (ABI)



Ankle-Brachial Index (ABI) is a ratio of ankle pressure to systemic (brachial) pressure.

BP cuff is inflated until pulse becomes inaudible via doppler

Perform at **posterior tibial, dorsalis pedis and brachial arteries**

Use the **highest** brachial pressure overall (systolic), and the **highest** ankle pressure per limb.

TABLE 62-4 Calculation of Ankle-Brachial Index

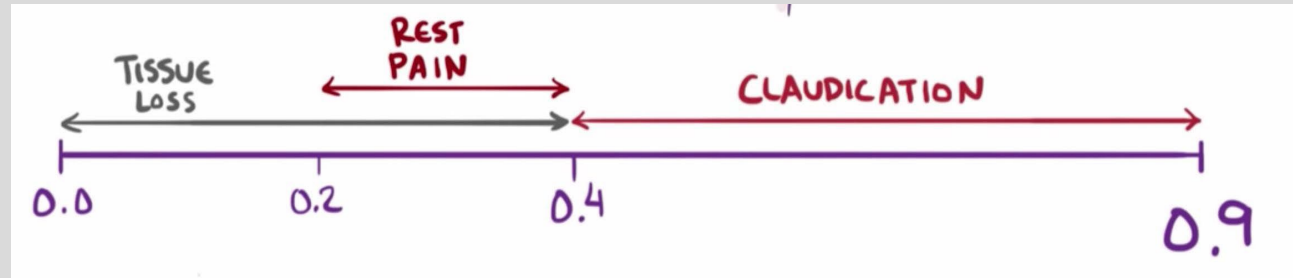
PARAMETER	RIGHT	LEFT
Brachial blood pressure	150 mm Hg	100 mm Hg
Dorsalis pedis	50 mm Hg	25 mm Hg
Posterior tibia	25 mm Hg	50 mm Hg



Diagnosis of PAD – Ankle-Brachial Index (ABI)

ABI of <0.9 is diagnostic for PAD

- In cases of vascular trauma or emergency, an ABI <0.9 indicates further diagnostic imaging (usually CT- or MR-angiography)
- ABI <0.4 is typically seen in patients with ischemic rest pain.
- Tissue loss occurs in patients with ABI <0.4 (these numbers vary)
- An ABI >1.3 occurs in patients with calcified arteries = non-valid!
 - Tends to occur in diabetics with ESRD



Normal = 1.0-1.2



PAD – Further Imaging

Digital subtraction angiography (DSA) remains the **gold standard**, however it is invasive, requiring arterial puncture. We reserve DSA when an intervention (i.e. stent or balloon angioplasty) is planned

What does “digital subtraction” mean?

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PAD – Further Imaging

CT-angiography (CTA) and **MR-angiography (MRA)** are non-invasive imaging studies widely used to identify aneurysms, dissections, stenoses, and occlusions.

What differentiates typical CT and MRI studies from CT- and MR- angiography?



CTA

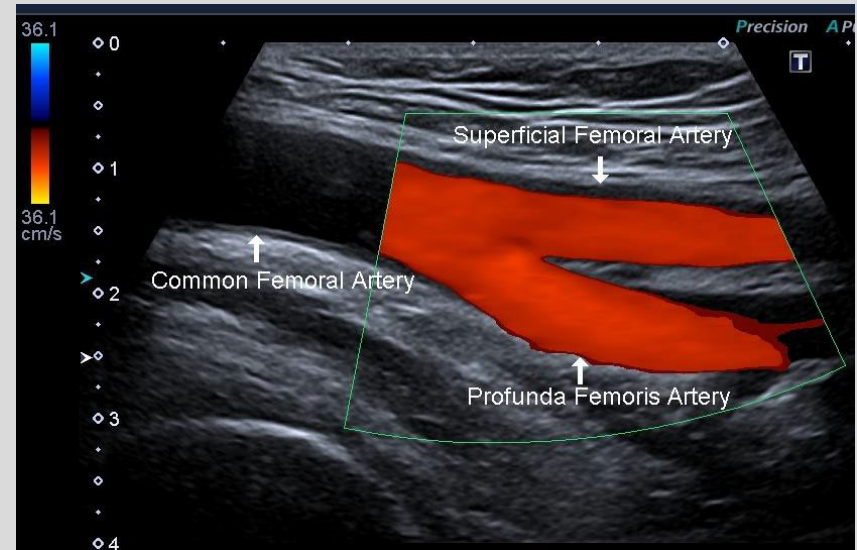
MRA

PAD – Further Imaging

Duplex ultrasonography (DUS) combines Doppler analysis and anatomic (B-mode) information to provide information on **blood flow velocity** and **vessel morphology**

Qualitative information includes **plaque morphology & turbulent flow**

Quantitative information used for diagnosis includes **peak systolic velocity (PSV)** and **end diastolic velocity (EDV)**



PAD – Complications

Acute Limb Ischemia (ALI)

Acute thromboembolic occlusion of the extremity, typically from **cardiac origin**, that most often lodges at a site of a **pre-existing plaque**.

Clinical Presentation (6 P's)

Pulselessness

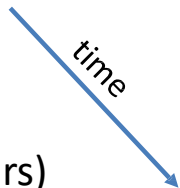
Pallor

Pain

Paresthesia

Paralysis (~6 hours)

Poikilothermia (coolness)



Chronic Limb-Threatening Ischemia (CLTI)

1-2% of patients with PAD

Clinical Presentation

Rest pain (ABI <0.4)

Tissue loss (ulcers, gangrene etc...)

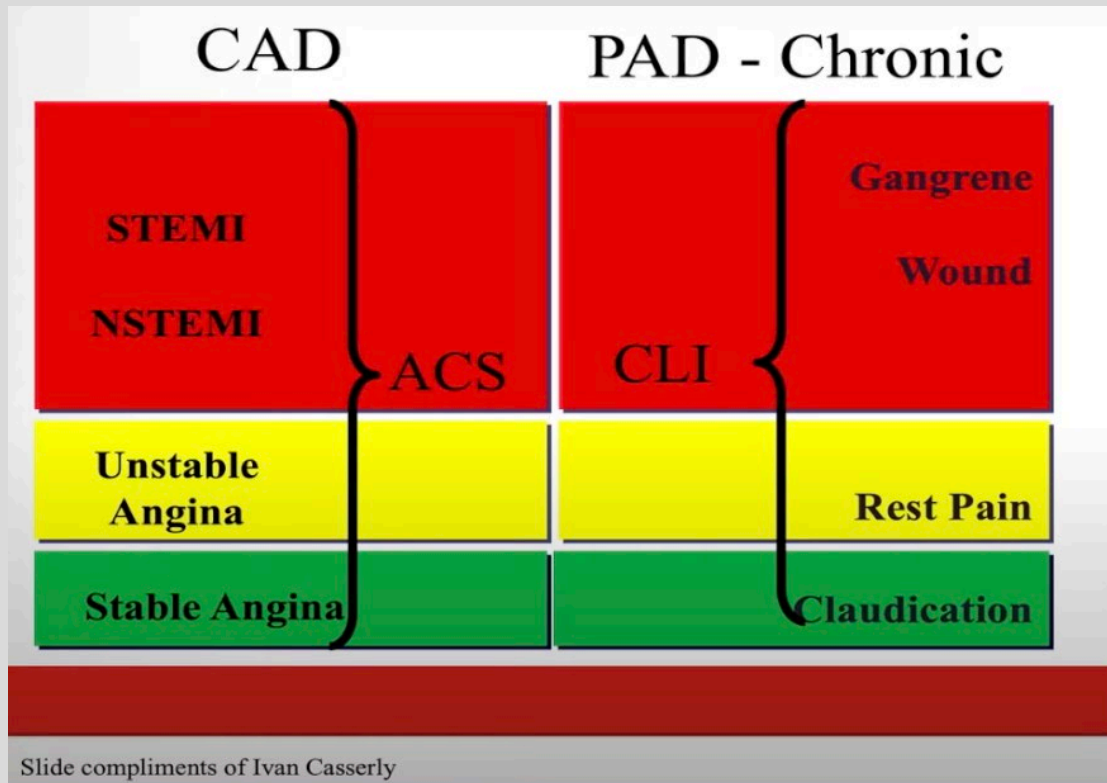
1-Year Prognosis

- 50% alive with both limbs
- 25% have amputation
- 25% CV-related mortality





PAD vs CAD





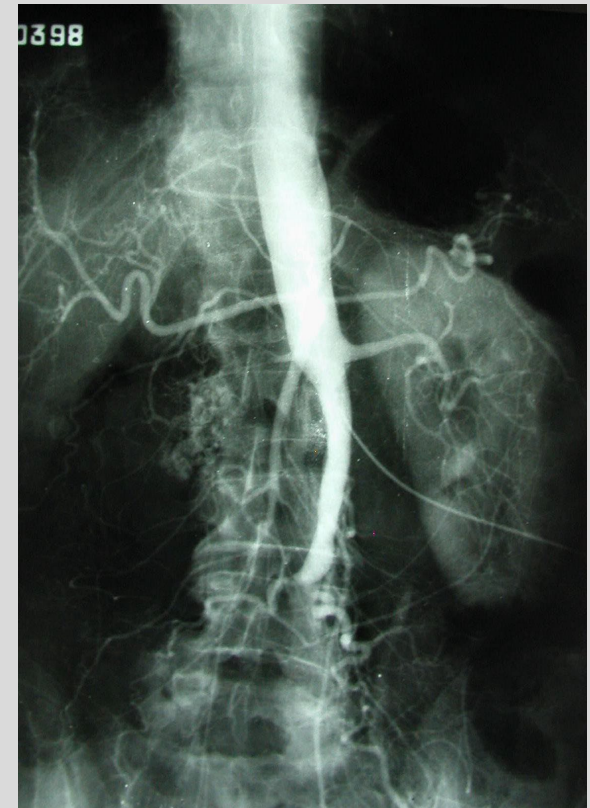
PAD – Leriche Syndrome

Aortoiliac Occlusive Disease – Affects the distal aorta, iliac arteries +/- femoropopliteal vessels

Clinical Presentation – Triad

- Buttock & thigh claudication
- Decreased femoral pulses
- Erectile dysfunction

Always keep **Leriche syndrome** on your differential diagnosis of erectile dysfunction and hip/thigh pain!



PAD – Treatment Options

Conservative Treatment (ABI>0.4, no CLTI [rest pain, tissue loss])

- Lifestyle changes: Smoking cessation, exercise, weight loss
- Medication: Statin, ASA, cilostazol (PDE3 inhibitor), blood glucose & BP control

Note: Vasodilators do not halt disease progression, just symptom relief

PAD patients are more likely to die of an MI or stroke than lose a limb!

>70% stenosis and/or **presence of CLTI** are *general* indications for open surgery (i.e. bypass of affected artery) or endovascular intervention (e.g. balloon dilation, stenting, atherectomy). Medical treatment is continued, along with constant wound care.

ALI is treated aggressively with anticoagulation (heparin), and subsequent imaging determines which intervention can be utilized.

Thrombolytics (tPA) + mechanical thrombectomy often used.

Question Break

A 68-year-old female presents to the clinic because of increased right upper calf pain when walking. She used to be able to walk a mile to work each morning but recently has noticed that at the half-way point, she must stop for a few minutes to “let her legs rest”. This happens every morning at the same distance from her house. She has a PMHx of HTN, DM and had a STEMI 3 years ago where she had a coronary stent placed in her LAD. She has been smoking a pack per day for the last 30 years and has no intention of quitting. She is currently taking Plavix, simvastatin, lisinopril and metformin. Physical exam shows a non-healing wound on her right lower leg.

What is the first step in the diagnosis of this patient?

At what artery is her blockage most likely?

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What is the first step in the diagnosis of this patient? Meticulous physical exam and perform an ABI

At what artery is her blockage most likely? Popliteal artery

Question Break

A 72-year-old female presents to the emergency department after 3 hours of excruciating left lower extremity pain. She reports no inciting events or recent trauma. Past medical history is relevant for coronary artery disease, treated with a 4-vessel CABG in 2012, atrial fibrillation, type-2 diabetes mellitus, hyperlipidemia, and COPD. On physical exam, you note her left leg is paler in color and cooler to the touch, when compared to the right. She has +2 femoral, popliteal, posterior tibial, and dorsalis pedis pulses on the right, but you are only able to appreciate a +2 femoral pulse on the left.

What is the most likely etiology of this patient's current condition?



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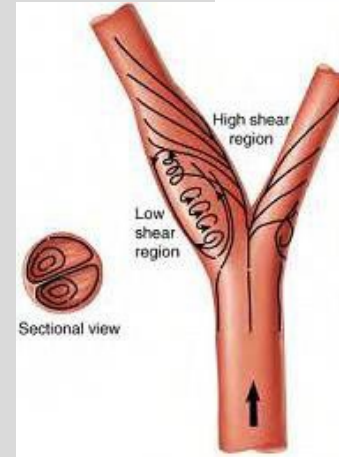
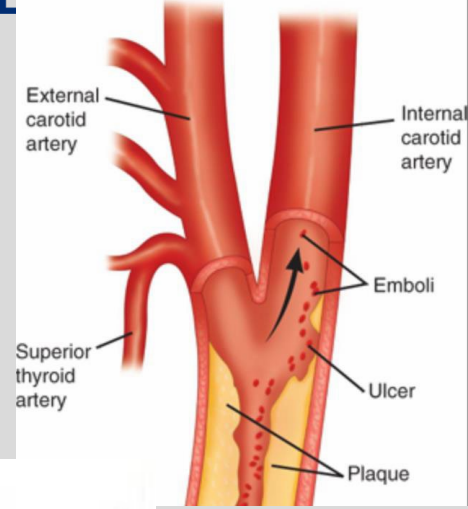
Atherosclerosis



Carotid Stenosis – Anatomy & Etiology

- Stroke is the 3rd leading cause of death and the leading cause of long-term disability in the US
- The most common cause of stroke is a **carotid atheroembolus**.
- The most common location for carotid plaque to form is the **carotid bifurcation**, due to **low shear stress**, which promotes plaque formation.
- The **first branch of the internal carotid artery (ICA) is the ophthalmic artery**. The ICA has no extra-cranial branches.

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Carotid Stenosis – Anatomy & Etiology

Transient Ischemic Attack (TIA)

- Acute (<24 hr) loss of focal cerebral function
 - Focal = correlates to one vascular supply of the brain (typically MCA).
Presents with localized symptoms
- Most are brief, lasting 2-15 minutes and rapid in onset
- Typically unilateral motor & sensory loss, aphasia, or dysarthria

Transient Monocular Blindness (TMB) aka amaurosis fugax

- Transient loss of unilateral vision due to ophthalmic a. occlusion
- “A shade came over my eye”
 - Location of vision loss correlates with occlusion of retinal artery branch
- Fundoscopy: Hollenhorst plaque (carotid atheroemboli)

Physical Examination

- Carotid bruit
- Screening test = Duplex ultrasound

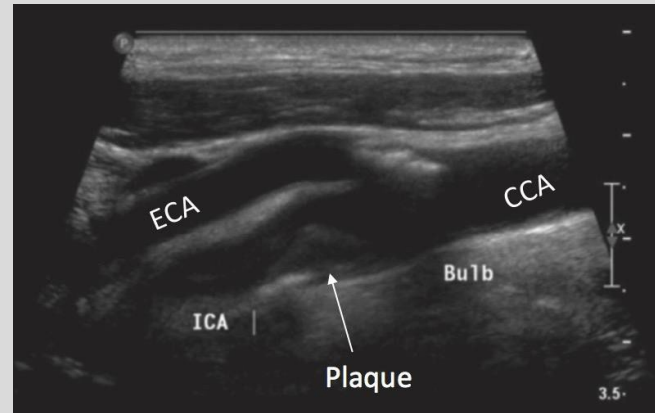


Question Break

A 79-year old female comes to your office for a follow-up visit to receive the results of her screening carotid duplex ultrasonography. Results show a marked 85% stenosis of the right internal carotid artery (pictured) and a minor 30% stenosis of the left carotid artery. She has been asymptomatic until this time.

If this patient were to experience a TIA, on which side of the body would you expect her neurologic symptoms to present?

If this patient were to experience TMB, which eye would most likely be affected?

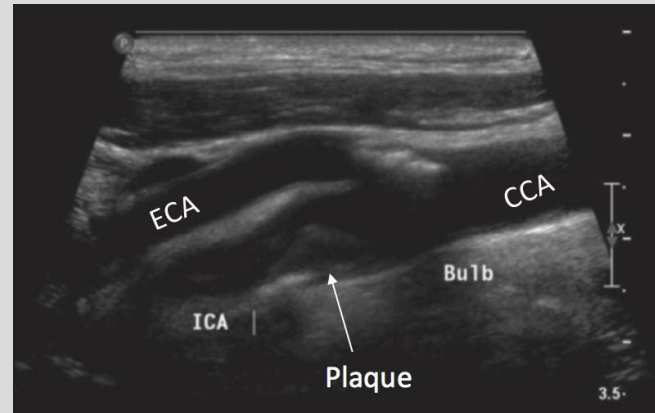


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If this patient were to experience a TIA, on which side of the body would you expect her neurologic symptoms to present? **Left**

If this patient were to experience TMB, which eye would most likely be affected? **Right**





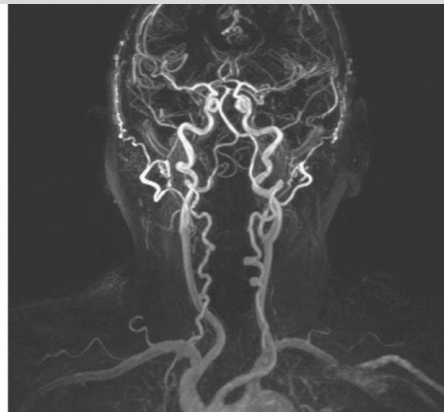
Carotid Stenosis – Diagnosis

Routine screening & first imaging modality of choice = **Duplex U/S**

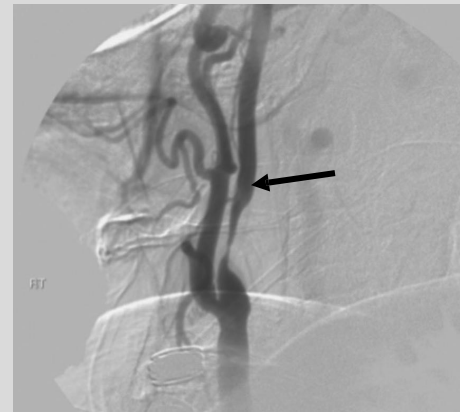
- Duplex only provides an **estimate** of stenosis based on **PSV**
- Follow-up tests include MRA or CTA
 - These modalities can also image vertebrobasilar & cerebral arteries
- Gold standard remains DSA, but is rarely done for diagnosis
 - Carotid stenting utilizes DSA intra-operatively for placement



CTA w/ 3D reconstruction



MRA w/ cerebral vessels



Digital subtraction angiography



Carotid Stenosis – Treatment Options

Treatment is determined based on presence of symptoms and degree of stenosis. In general, **any stenosis indicates need for antiplatelet (ASA +/- clopidogrel) and statin therapy**. Indications for surgical procedures is not important to know at this point.

Carotid Endarterectomy (CEA)

- Open surgery with great results, where carotid artery is entered via incision (arteriotomy), and plaque is removed (atherectomy)
- Risk of cranial nerve injury (CNs IX, X, XI, & XII) & stroke

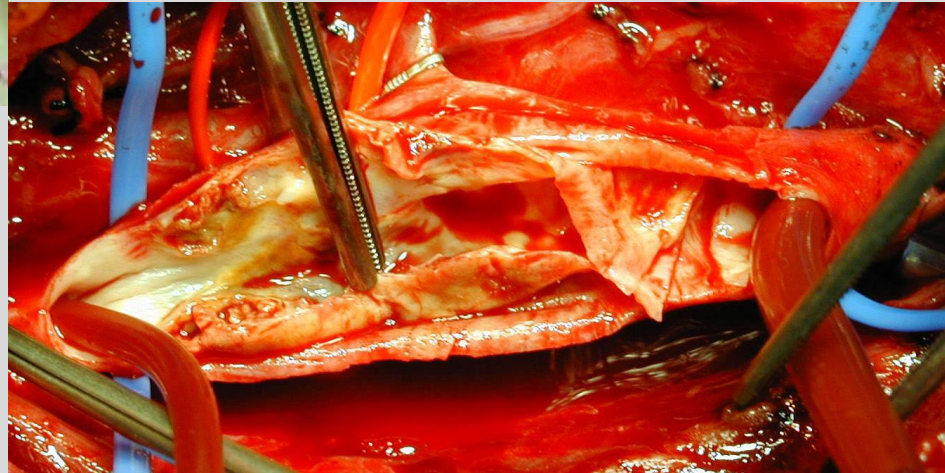
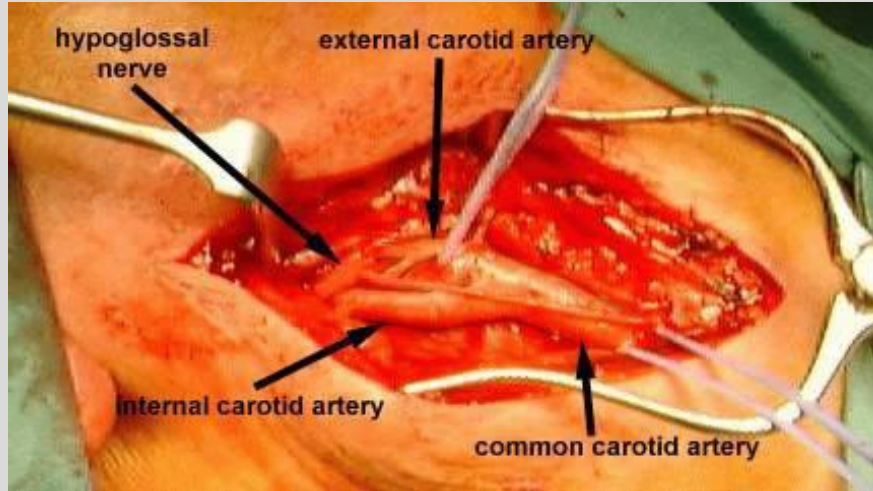
Carotid Stenting (CAS)/Trans-Carotid Revascularization (TCAR)

- Minimally invasive techniques for patients unable to tolerate open surgery
- Stent is placed via femoral access (CAS) or carotid access (TCAR)
- CAS = controversial due to poor outcomes when compared to CEA
- TCAR = new “hybrid” technique involving localized stent placement

Landmark clinical trials for CEA vs CAS: CREST & CREST-2 (ongoing)

Carotid Stenosis – Carotid Endarterectomy

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Venous Disease

Chronic Venous Insufficiency (CVI)

- Highly variable spectrum of disease involving the lower extremity
- Most common vascular disorder in the US at ~30% of population
- Estimated \$1B annual burden to the US healthcare system
 - Highly misunderstood, under-recognized, & impact under-estimated



Chronic Venous Insufficiency (CVI) – Anatomy

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Superficial Veins

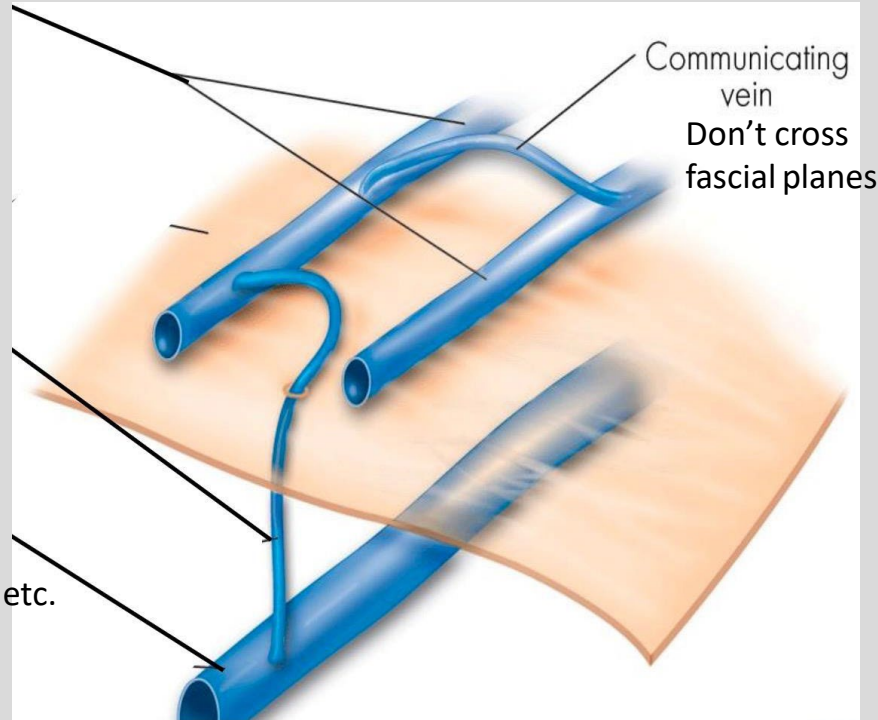
GSV, SSV, ASV

Perforator Veins

Cross fascial planes
Superficial → Deep
One-way valves

Deep Veins

Femoral, Popliteal, etc.



Chronic Venous Insufficiency (CVI) – Anatomy

Deep Veins

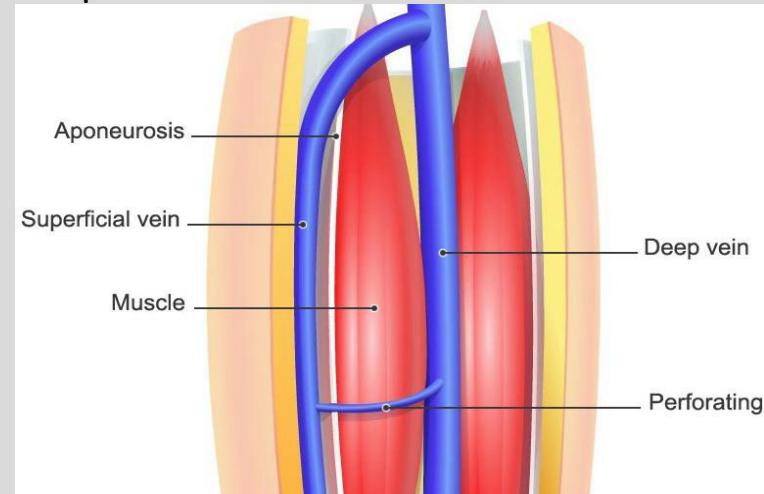
- Femoral, popliteal, peroneal etc. lie within musculature
- Calf muscles act as a pump to return venous blood to the heart

Superficial Veins

- Saphenous veins lie close to the skin
- Lack of surrounding musculature → must rely on valves to prevent backflow

Venous Valves

- Unidirectional → prevent backflow
- Density varies → highly concentrated in the distal extremities



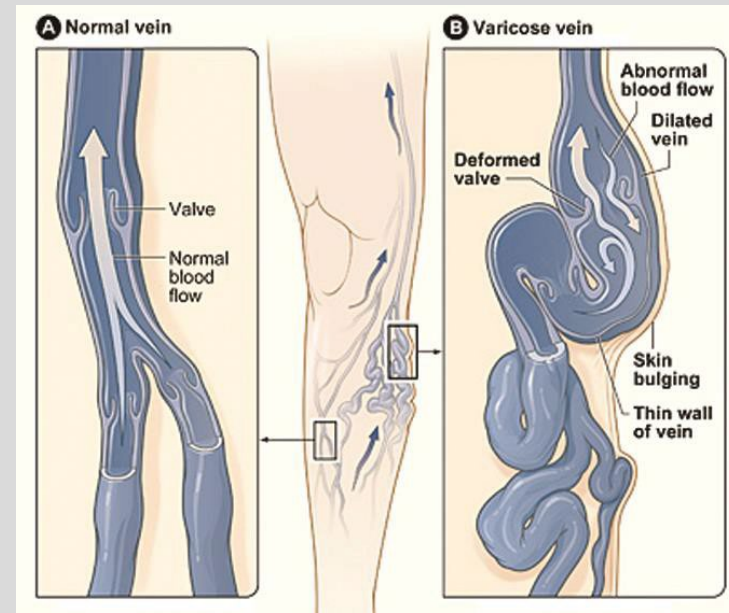
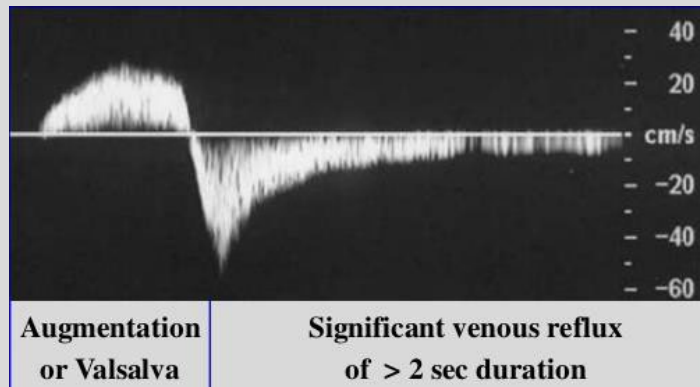


CVI – Pathophysiology

While the pathophysiology is still not completely elucidated, **venous valve incompetence** is the primary underlying mechanism of CVI. Reflux causes stasis and pooling of blood in leg.

Best test: Duplex ultrasonography

- Reveals dilated veins & reflux

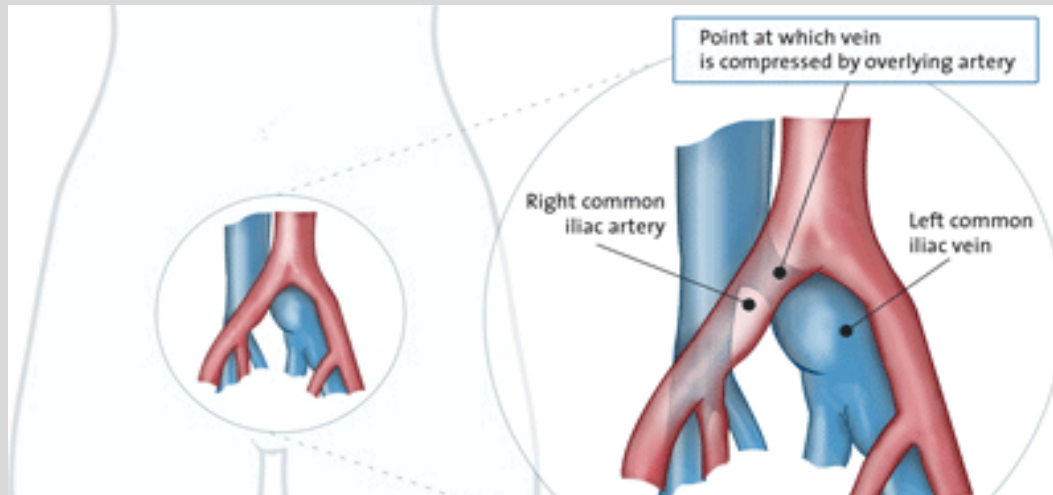




CVI – Pathophysiology

May-Thurner Syndrome (MTS) → compression of the left common iliac vein by the right common iliac artery

- Edema of the entire (or most of the) extremity
- Deep venous thrombosis (DVT)
- Venous claudication (leg pain/tightness with exercise)
- Varicosities associated with valve incompetence > MTS
- Treat outflow obstruction with vein stenting



CVI – Pathophysiology

CVI can be primary or secondary in nature

- **Primary** → due to anatomic defects in veins or valves
- **Secondary** → post-thrombotic syndrome (PTS) following DVT, leads to venous occlusion and inflammation that damages lumens & valves

Natural History of CVI



Varicose veins
Valves cannot handle elevated pressure → reflux



→ **Edema**
Extravasation of RBCs into the interstitium



→ **Pigmentation**
Skin breakdown → infection of exposed tissue or blood within subQ space



→ **Ulceration**
Mechanical wall stress → release of inflammatory mediators

Venous ulcers are most commonly found 2-3 cm above medial malleolus

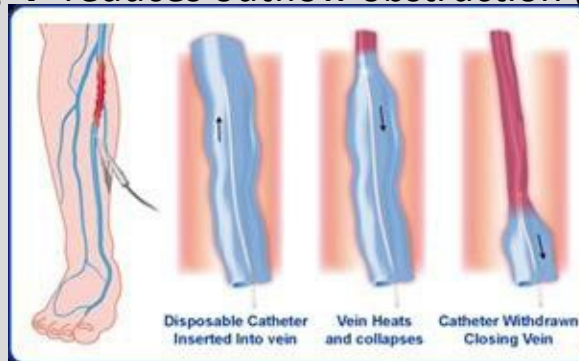
CVI – Treatment

Conservative Therapy

- Compression stockings → external pressure assists venous return
- NSAIDs for pain relief

Interventions

- Sclerotherapy → injected chemicals or foam to shrink varicosities
- Endovenous ablation → laser or radiofrequency signals create heat, which closes vein → fibrosis leads to permanent vein closure
 - Venous blood is then re-routed to more competent & deeper veins
- Iliac vein stenting → reduces outflow obstruction (i.e. MTS)



Endovenous Ablation



Sclerotherapy

Question Break

A 38-year-old woman presents to your attending vascular surgeon's clinic and you are the first to see her. She complains of "gross veins popping out of her leg", swelling, and pain while standing at her job as a barista. The pain and edema worsen as the day progresses. On physical exam you note an overweight (BMI = 29.6) female with multiple, 2- and 3-cm varicose veins of the left and right lower extremity that are compressible and non-tender.

Duplex ultrasonography reveals compression of left common iliac vein.

What is the most likely etiology of this venous compression?

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May-Thurner Syndrome

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Arterial vs Venous Ulcers – Comparison

Characteristic	Arterial Ulcer	Venous Ulcer
History	Smoking, PAD, CAD, diabetes, previous vascular disease.	DVT, varicose veins, obesity, trauma, surgery, pregnancy.
Location	Distal extremities (i.e. toes, fingers) & bony prominences.	Between lower calf and medial malleolus.
Ulcer Bed Ulcer Borders	Dry, necrotic tissue. Regular, punched-out margins.	Fibrous material with exudate. Irregular margins.
Condition of Extremity & Pulse Exam	Thin, shiny, hairless skin. Pallor, reduced temperature. Diminished pulses & capillary refill.	Brawny, thick, itchy & fibrotic skin. Pigmentation with edema. Normal pulse & capillary refill.
Treatment	Revascularization, anti-platelets, statins, & risk-factor modification (e.g. smoking, blood-glucose).	Compression therapy, leg elevation, endovenous ablation (especially of perforator veins), iliac vein stenting.



Lymphatic System – Pathology & Clinical Presentation

The lymphatic system is a highly porous, low pressure capillary system. Has a primary immunologic role by **filtering proteins and cells**. In the GI system, lymphatics absorb large fat cells into the blood stream.

Lymphedema occurs when the lymphatic load exceeds the system's carrying capacity, either due to excess lymph production, or **obstruction of lymphatic outflow**.

Lymphedema stimulates **fat deposition** → ~50% of lymphedema volume is fat (does not pit)!

Primary Lymphedema

- Due to anatomic defect
 - Milroy disease
 - *Lymphedema praecox*
 - Marked dorsal foot swelling
 - Hypoplastic toenails

Secondary Lymphedema

- Due to:
 - Infection (*W. bancrofti*)
 - Trauma
 - Prior surgery
 - Lymph node dissection
 - Radiation therapy
 - Cancer



Lymphedema – Staging & Complications

Classification System

Stage I → Completely relieved by elevation → pitting

Stage II → Incompletely relieved by elevation → variable pitting

Stage III → Elephantiasis with severe fibrosis, warty overgrowths, and **fat deposition** → *no pitting*

Lymphangiosarcoma

- Malignant cancer secondary to long-standing lymphedema
 - Consider in patients with skin changes
 - Stewart-Treves → following lymph node dissection with mastectomy
- Most common in upper extremity
- Treatment = limb amputation



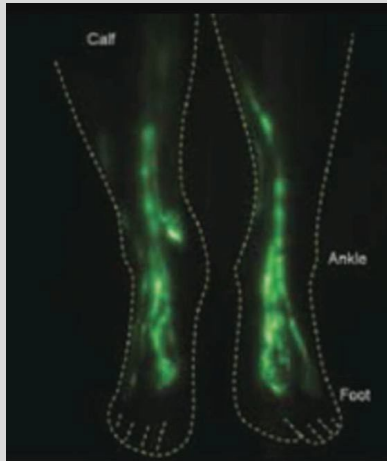
Lymphedema – Diagnosis & Imaging

Diagnosis can often be made based on **history** and **physical** alone.

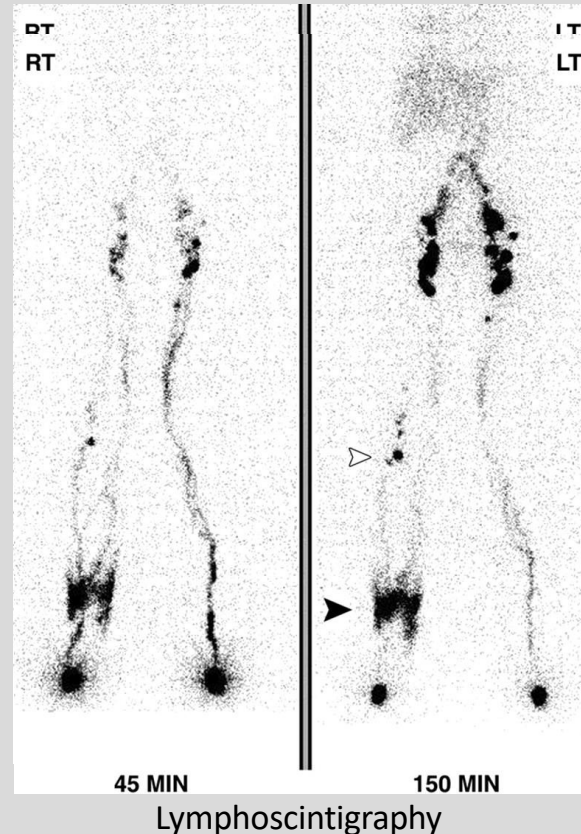
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Imaging

- Lymphoscintigraphy
- Indocyanine green (ICG) lymphography



ICG lymphography



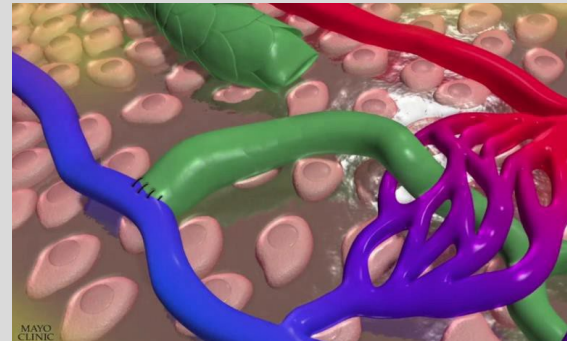
Lymphedema – Management & Treatment

Conservative Therapies

- Compression or velcro stockings
- Exercise
- Pneumatic compression devices
- Lymphatic OMT
- **Gold standard** = manual decompressive therapy

Surgical Interventions

- Lymphovenous bypass
- Vascularized lymph node transfer
- Excisional debulking
- Liposuction → remember, lymphedema contains a lot of adipose tissue!



Lymphovenous bypass

Question Break



While on your internal medicine rotation at Queens Hospital Center, a 56-year-old man who recently arrived from Bangladesh is admitted to your team's service. He complains of progressive limb swelling over the past 3 years. He works as a farmer and has not seen a doctor in 2 decades. He reports a past medical history of diabetes and high cholesterol. He smokes 1.5 packs of cigarettes per day, and has been smoking since he was a teenager. On physical exam you notice a markedly enlarged right lower limb that is firm and tough to the touch. Elevation of the leg does not relieve the swelling. A blood smear is collected and Giemsa staining reveals a long, cylindrical worm.

What is the most likely inciting factor of his disease process?

Question Break



While on your internal medicine rotation at Queens Hospital Center, a 56-year-old man who recently arrived from Bangladesh is admitted to your team's service. He complains of progressive limb swelling over the past 3 years. He works as a farmer and has not seen a doctor in 2 decades. He reports a past medical history of diabetes and high cholesterol. He smokes 1.5 packs of cigarettes per day, and has been smoking since he was a teenager. On physical exam you notice a markedly enlarged right lower limb that is firm and tough to the touch. Elevation of the leg does not relieve the swelling. A blood smear is collected and Giemsa staining reveals a long, cylindrical worm.

What is the most likely inciting factor of his disease process? *W. bancrofti*

Overview

Approach to the Vascular Patient

- History, physical (inspect, palpate, auscultate, & ultrasound)
- Meticulously document all physical exam findings and pulse exam (pictures included)
- Imaging: U/S, CT- & MR-angiography, digital subtraction angiography (DSA)

Arterial Disease

- Peripheral Arterial Disease → atherosclerosis of lower extremity
 - Claudication: pain with activity, relieves with rest, reproducible
 - Leriche syndrome, aortoiliac occlusion: claudication, decreased pulses, & erectile dysfxn
 - ALL: acute thromboembolic occlusion at previously stenotic site
 - 6 P's: pain, pallor, pulseless, paresthesia, paralysis, poikilothermia
 - CLTI: rest pain, tissue loss with high-risk of morbidity & mortality
 - ABI = fraction of highest ankle pressure / highest brachial pressure
 - <0.9 = PAD; <0.4 = rest pain; <0.4 = tissue loss
- Carotid Stenosis → atherosclerosis of bifurcation; leading cause of stroke
 - Presents with contralateral TIA and/or ipsilateral amaurosis fugax
 - Screen with duplex U/S, follow-up with CTA or MRA

Chronic Venous Insufficiency

- Superficial, deep, and perforator veins create network of return outflow
- Spectrum of disease from telangectasias to venous stasis ulcers
- Venous incompetence → valve dysfunction → superficial reflux
 - Diagnose with duplex U/S, which shows venous dilation & reflux
 - Treat with compression therapy → ablation/sclerotherapy
- May-Thurner Syndrome (MTS) → compression of L. CIV by R. CIA
 - Treat with compression therapy → iliac vein stenting

Lymphedema

- Primary: congenital, due to anatomic deformities
- Secondary: *W. bancrofti* infection, lymph node dissection, trauma, surgery
- Lymphangiosarcoma: cancer secondary to long-standing lymphedema
- Treat with compression therapy or surgery

Thank you!!

For any questions, please feel free to email me at

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Lecture Feedback :

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>